WHEN THE SEASONS CHANGE: FEDERATED LEARNING WITH CYCLIC DRIFT AND MEMORY CONSTRAINTS

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ABSTRACT

We introduce *Cyclic Drift Federated Learning with Finite Memory (CDFL-FM)*, a framework for federated continual learning. Each client experiences a recurrent Markovian data arrival process and maintains a bounded local buffer to capture streaming environment with cyclically evolving data distributions, motivated by recurring concept drift in practical applications. At each discrete time instant, U clients receive a new data batch, stored in finite memory by replacing the oldest batch. With some probability, the incoming batch may be from the current environment distribution (fresh data) or the previous one (stale data). Clients use their local buffer to compute gradient updates for a centralized model, as in standard federated learning. Due to complexity limitations, the parameter server (PS) can only collect updates from a subset of clients. We propose COS4-SELECT, a client selection mechanism in which each client reports a single scalar quantifying the discrepancy between its local and the global gradient. In particular, COS4-SELECT is based on a cosine similarity under the L_4 norm, which we identify as a sensitive and lightweight measure of "freshness" in the presence of cyclic drift and finite memory. Numerical simulations show that this enables sensitive detection of distribution shifts and improves model performance under CDFL-FM.

Index Terms— Federated Learning; Concept Drift; Client Selection.

1. INTRODUCTION

Consider a scenario in which multiple remote observers collect data from a shared environment subject to cyclic changes. Each observer experiences these changes asynchronously: with some probability a new observation reflects the current distribution, while with the complementary probability it reflects the past distribution. Because each client maintains only a finite local memory, older samples are gradually discarded, so the available data reflects a sliding window of recent observations. A central parameter server (PS) seeks to maintain a global model that remains accurate across these evolving distributions. Since the number of clients is large and communication resources are constrained, it is impractical for the PS to aggregate updates from all clients at every round. To address this challenge, we allow each client to transmit a compact summary of its local gradient [1]. The central question then becomes: *how should the PS use these summaries to identify the most informative subset of clients, those best positioned to capture distributional shifts in the environment?* In this paper, we argue that the finite-memory constraint amplifies the importance of timely and accurate client selection: informative updates may be lost if clients are not promptly chosen. A carefully designed measure of "freshness" quantifying the discrepancy between local and global gradients allows the PS to prioritize clients carrying new information, thereby improving the performance of the global model in tracking the updated environment distribution.

The system model in this paper lies at the intersection of three themes: (i) cyclic concept drift in continual learning, (ii) federated learning under continual learning, and (iii) client selection in federated learning. We next review the literature related to each of these topics.

Cyclic concept drift: Concept drift denotes changes in the data distribution over time. In the case of *cyclic* drift, the distribution revisits earlier regimes, motivating algorithms that can detect recurrence and reuse prior knowledge rather than relearn from scratch [2, 3]. Typical approaches include error or distribution-based drift detectors, ensemble or rehearsal mechanisms, and lightweight online updates that balance stability and plasticity. In continual learning, cyclic drift exacerbates *catastrophic forgetting* [4], which has been addressed through replay buffers, regularization based constraints, or modular architectures [5–7].

Federated continual learning: Federated continual learning combines decentralized model training with non-stationary data streams. This coupling introduces heterogeneity and asynchrony across clients, catastrophic forgetting at multiple levels, and limited resources and communication. Recent surveys [8, 9] categorize approaches into (i) memory- or prototype-based replay, (ii) parameter regularization [10, 11], and (iii) knowledge transfer via distillation or ensembling [12].

Client selection in federated learning: Client selection is

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crucial for federated learning, as only a subset of devices can participate each round. In non-stationary settings, uniform sampling may overlook distributional shifts, while infrequent participation prevents some clients from adapting. To mitigate this, recent studies propose drift-aware strategies that prioritize clients based on the novelty or recurrence of their data [13].

1.1. Contributions

The main contribution of this work is the formulation of a federated learning framework, CDFL-FM, that explicitly couples cyclic environment drift with finite client memory, thereby inducing latent heterogeneity in the local datasets. Within this model, we benchmark several lightweight client selection policies. Our results demonstrate that similarity feedback based feedback is surprisingly effective: the $\cos_2$ policy achieves performance close to the arrival aware baseline without requiring access to arrival information, while the $\cos_4$ policy reveals the sensitivity of similarity metrics to class complexity and feature diversity.

Notation: Calligraphic letters denote sets. Bold lowercase letters (e.g., $\mathbf{x}$) represent vectors when the dimension is clear. Random variables are denoted by uppercase letters, and their realizations by lowercase. $\mathbf{a} \mid \mathbf{b}$ denotes the concatenation of the vectors $\mathbf{a}$ and $\mathbf{b}$. Finally, the set $\{0, 1, \ldots, n-1\}$ is denoted by $[n]$.

2. SYSTEM MODEL: CYCLIC DRIFT FEDERATED LEARNING WITH FINITE MEMORY (CDFL-FM)

We consider U remote clients observing a environment whose data distribution evolves over time, denoted by $P_{\mathcal{D}}(t)$. The clients collaborate with a central PS to minimize the global loss

$$\mathcal{L}(\mathbf{w}(t)) \triangleq \frac{1}{|\mathcal{D}(t)|} \sum_{u \in [U]} \mathcal{L}_u(\mathcal{D}_u(t), \mathbf{w}(t)), \tag{1}$$

where $\mathcal{D}(t) \triangleq \biguplus_{u \in [U]} \mathcal{D}_u(t)$ is the aggregate dataset and $\mathcal{D}_u(t)$ is the local dataset of client u at time t.

At each time $t \in \mathbb{Z}$, the system proceeds through the following phases, also illustrated in Fig. 1.

Phase 0: Model distribution. The PS broadcasts a global update $\mathbf{g}(t)$ to all clients. By accumulating these updates, each client reconstructs the current global model $\mathbf{w}(t)$.

Phase 1: Data observation. Client u is provided with a new batch of B observation label pairs, consisting of observations $\mathbf{x}_b$ and corresponding labels y_b, from the environment

$$\mathcal{D}_u(t) = \{(\mathbf{x}_b, y_b)\}_{b \in [B]}. \tag{2}$$

The environment distribution evolves cyclically across K phases of length N:

$$P_{\mathcal{D}}(t) = \lfloor t/N \rfloor \bmod K. \tag{3}$$

Each client observes current data with probability λ, or stale data from the previous phase with probability $1 - \lambda$:

$$\mathcal{D}_u(t) \sim \begin{cases} P_{\mathcal{D}}(\lfloor t/N \rfloor \bmod K), & \text{w.p. } \lambda, \\ P_{\mathcal{D}}((\lfloor t/N \rfloor \bmod K) - 1), & \text{w.p. } 1 - \lambda. \end{cases} \tag{4}$$

We assume that neither the PS nor the user are aware of the environment distribution or its latent state.

Phase 2: Memory update, local training, and gradient summaries. Each client maintains a FIFO buffer of size M batches,

$$\mathcal{C}_u(t) = [\mathcal{D}_u(t - M + 1) \mid \ldots \mid \mathcal{D}_u(t)], \tag{5}$$

where $\mid$ denotes concatenation. The pair of (4) and (5) induces a cyclic Markov chain for $\mathcal{C}_u(t)$. From $\mathcal{C}_u(t)$, client u computes a stochastic update via SGD:

$$\mathbf{g}_u(t) \approx \nabla \mathcal{L}(\mathcal{C}_u(t), \mathbf{w}(t)). \tag{6}$$

Instead of transmitting full updates, a client reports a scalar summary

$$q_u(t) = f_q(\mathbf{g}(t-1), \mathbf{g}_u(t)), \tag{7}$$

which is lightweight compared to $\mathbf{g}_u(t)$.

Phase 3: Client selection. Based on the scalar summary obtained from the previous phase, PS will decide which client is eligible for transmitting by

$$\mathcal{S}(t) = \pi(\mathbf{q}(t)), \qquad |\mathcal{S}(t)| = S. \tag{8}$$

Phase 4: Aggregation. The selected clients report their gradients, used by the PS to compute the next model update.

$$\mathbf{g}_n = \frac{1}{S} \sum_s \mathbf{g}_n^{(s)}, \tag{9}$$

Phase 5: Environment drift. The environment transitions to the next state, i.e., from $P_{\mathcal{D}}(t)$ to $P_{\mathcal{D}}(t + kN)$.

2.1. Objective

The design goal is to select both the summary function f_q and the selection policy π such that the global model tracks the evolving optimum. Let

$$\mathbf{w}_k^* = \operatorname*{argmin}_{\mathbf{w}} \ \mathbb{E}_{\mathcal{D} \sim P_{\mathcal{D}}(k)} \mathcal{L}(\mathbf{w}, \mathcal{D}), \tag{10}$$

with minimum loss L_k^*. We measure deviation from the ideal trajectory as

$$\min_{f_q, \pi} \sum_t \Delta_t, \qquad \Delta_t = L_k^* - \mathbb{E}[\mathcal{L}(t, \mathbf{w}(t))], \tag{11}$$

where the expectation is over randomness in data arrivals (4), memory updates (5), and initialization.

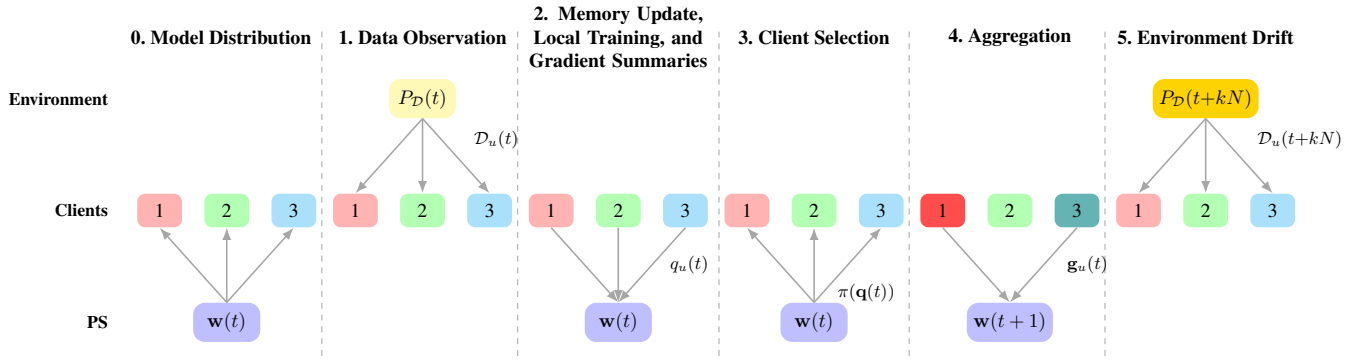

Fig. 1: Workflow phases for model distribution, data distribution, user training, client selection, aggregation, and environment drift. The environment state shifts every N timeframes.

Table 1: A summary of the notation introduced in Sec. 2.

quantity	notation
number of clients	U
selected client each timeframe	S
number of timeframes	T
number of timeframes per cycle	N
number of memory cells per client	M
size of each memory cells	B
content per memory cells	$\mathcal{C}$
number of environment states (classes)	K
gradient of client u at timeframes n	$\mathbf{g}_n^{(u)}$
gradient of PS at timeframes n	$\mathbf{g}_n$
dataset per client	$\mathcal{D}^{(u)}$
arrival probability	λ

This model highlights three intertwined challenges in distributed learning: (i) *cyclic concept drift*, from the periodic environment distribution; (ii) *finite client memory*, which forces forgetting of outdated data; and (iii) *communication constraints*, motivating scalar feedback and selective participation. Together, these elements define the CDFL-FM framework. A summary of the notation introduced in this section is provided in Table 1.

3. PROPOSED APPROACHES

In this section, we investigate a set of client selection metrics that are widely studied in the literature. Each policy specifies how the PS chooses S clients at each round based on lightweight statistics from the local updates. We focus on the following strategies:

- **Vanilla (Random).** Clients are selected uniformly at random from the pool of U users, without considering data heterogeneity. This serves as a neutral baseline.
- **Cosine-p Similarity.** Following [1], the similarity between the previous global gradient $\mathbf{g}(t-1)$ and the local update $\mathbf{g}_u(t)$ is measured as

$$\cos_p(\mathbf{x},\mathbf{y}) = \frac{\langle \mathbf{x},\mathbf{y}\rangle_p}{\|\mathbf{x}\|_p \, \|\mathbf{y}\|_p}, \tag{12}$$

  where $\langle \mathbf{u},\mathbf{v}\rangle_p = \frac{\|\mathbf{u}+\mathbf{v}\|_p - \|\mathbf{u}-\mathbf{v}\|_p}{p}$. and $p \in \{2,4\}$. Both *similarity* (selecting the most aligned clients) and *dissimilarity* (selecting the least aligned clients) are considered.
- **Age of Gradient (AoG).** Inspired by [14], each client maintains some knowledge of the gradients from the previous rounds. γ, the forgetting coefficient, is implemented to decide the rate of forgetting previous gradients.
- **Arrival-Aware (Genie-Aided).** Each client reports the number of newly arrived batches in its memory at round t. Clients with more fresh data are prioritized. This policy is idealized, as it assumes the server has direct access to the freshness of client memory.

4. NUMERICAL EVALUATIONS

Experimental Settings For CIFAR-10, we adopt a ResNet-18 backbone, while for MNIST we employ a lightweight 2-Conv-2-FC CNN. In both cases, only the dense (fully connected) layers are updated during local training, with convolutional layers frozen. The setups differ mainly in the number of communication rounds: CIFAR-10 is trained with $U = 10$ clients over $T = 225$ rounds, while MNIST uses $U = 10$ clients over $T = 150$ rounds. To simulate recurring distribution shifts, we define a *cycle* as the full sequence of drift across all K environment states before returning to the initial distribution within the duration of N timeframes. Both datasets ran for 3 cycles, corresponding to 75 rounds per cycle for CIFAR-10 and 50 rounds per cycle for MNIST. The drift is modeled as cyclic, with a memory size of $M = 4$ cells per client. The number of states K is defined by grouping the original dataset labels (e.g., pairing digits in MNIST or labels in CIFAR-10) into K meta-classes, each of which is introduced sequentially to simulate distributional shifts. Here both MNIST and CIFAR-10 has K = 5 environment states.

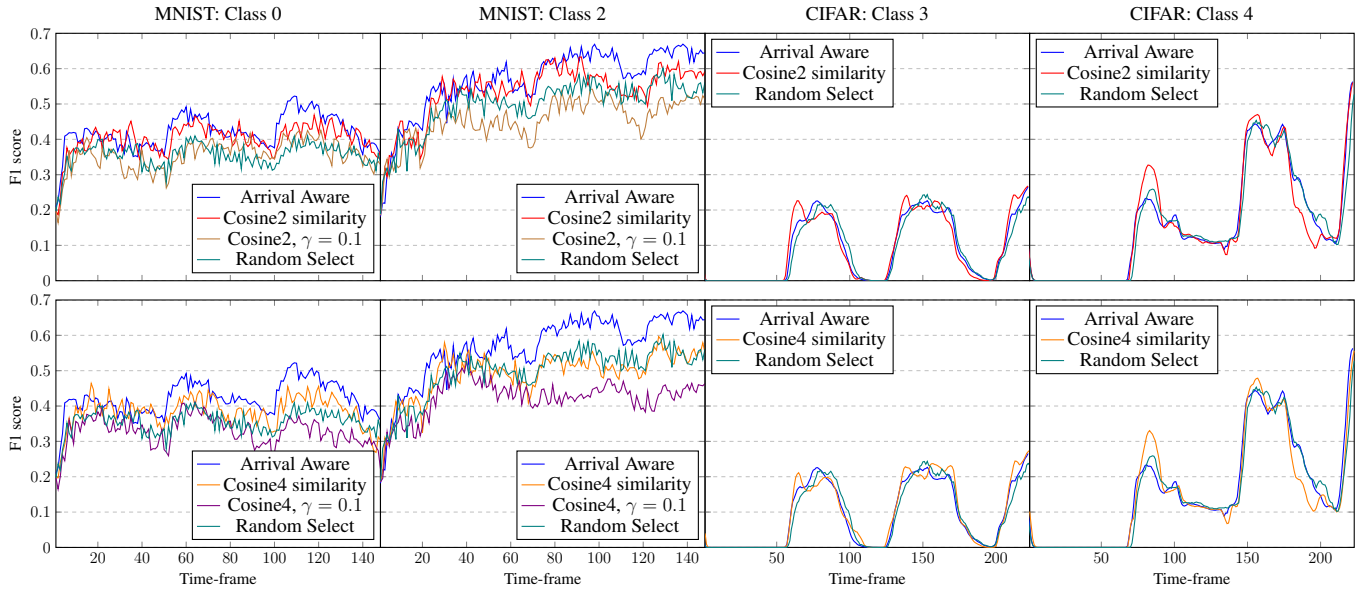

Fig. 2: Mean F1 score across the MNIST and CIFAR-10 datasets, with newly mapped classes: Class 0 and Class 2 in MNIST representing digits (0, 1) and (4, 5), respectively. Class 3 and Class 4 in CIFAR-10 corresponding to labels (cats, automobiles) and (deer, dogs), respectively. The results are presented under the $\cos_2$ and $\cos_4$ similarity feedback policies, compared against reference benchmarks (arrival aware and random selection). The experiment considers $K = 5$ environment states and goes through the phases described in Sec. 2.

Performance on MNIST The first row of plots in Fig. 2 presents results for the $\cos_2$ policy, while the second row displays those for the $\cos_4$ policy. From figure 2 the F1 score demonstrates that the arrival-aware policy maximizes precision recall balance with perfect foresight of which client provides the newest information, serving as an upper bound. The $\cos_2$ policy closely approximates this performance, often achieving comparable or slightly better F1 scores without such foresight, while the $\cos_4$ policy shows mixed results, excelling in Class 0 (labels 0 and 1) but underperforming in Class 2 (labels 4 and 5). Both freshness feedback policies generally outperform the random selection policy across most timeframes. The sharper gains in Class 2 suggest greater feature diversity (e.g., the distinct shapes of digits 4 and 5) compared to the simpler Class 0 (digits 0 and 1, with more uniform structures), indicating that the efficacy of these methods may depend on the complexity and variability within each class. We also introduced "Age of Gradients (AoG)" with forgetting coefficient $\gamma = 0.1$ to assist the $\cos_2$ similarity policy. From the numerical results using the MNIST dataset, it is shown that it has no positive impact. Instead, the AoG enhanced $\cos_p$ policy performs even worse than the random selection policy. This underperformance may stem from noise or misalignment in the cosine similarity metric due to aged gradients. Therefore, the AoG mechanism may not be needed under cyclic environment drift.

Performance on CIFAR-10 From Fig. 2 we can observe that for Class 3, both $\cos_2$ and $\cos_4$ policies exhibit the fastest rise to peak F1 scores during that cycle. Reaching over 0.2 within the first 60 timeframes. While it took the arrival aware and random policies around 80 timeframes to achieve the same F1 score. This rapid improvement suggests these similarity feedback based methods effectively prioritize clients with novel information early on. In Class 4, a similar pattern emerges, with policies $\cos_2$ and $\cos_4$ achieving a higher F1 score than arrival aware and random selection within approximately the same amount of time. These results reveal a tradeoff: $\cos_2$ and $\cos_4$ accelerate to higher early peaks without perfect foresight, often exceeding the arrival aware policy in speed and initial height, but may decline as data evolves due to potential oversensitivity to similarities.

5. CONCLUSION

In this work, the experimental results highlight the effectiveness of similarity feedback based client selection policies in federated learning under cyclic drift. The $\cos_2$ policy consistently provides a strong approximation of the arrival aware upper bound, while the $\cos_4$ policy exhibits class dependent behavior excelling when feature diversity is high but lagging otherwise. Moreover, the faster convergence of $\cos_2$ and $\cos_4$ in Classes 3 and 4 illustrates their ability to prioritize similar clients early, though at the cost of potential oversensitivity as the distribution evolves. These advancements promise to refine user selection techniques, paving the way for more effective federated learning systems in dynamic environments.

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